# Quant Research OS

## 量化研究操作系统

Technical Handbook & System Walkthrough

技术手册与系统详解（英中对照）

Architecture · Quant Theory · Orchestration · Engines · Workstation UI · Operations

架构 · 量化理论 · 编排 · 引擎 · 工作站界面 · 运维

| | |
|---|---|
| **Hard rule:** the planner/agent may propose research directions, but **deterministic quantitative engines are the sole source of truth** for financial metrics, positions, and risk numbers. | 硬性规则：规划器/智能体可以提出研究方向，但确定性量化引擎是财务指标、持仓与风险数字的唯一真相来源。智能体不得编造夏普比率等指标。 |

| | |
|---|---|
| Audience: quantitative researchers, platform engineers, and reviewers who need to understand how research questions become experiments, alphas, portfolio impact, and auditable reports. Layout: **English left / Chinese right**. | 读者：量化研究员、平台工程师与评审者——需要理解研究问题如何变成实验、Alpha、组合影响与可审计报告。版式：左侧英文 / 右侧中文。 |
| Chinese font embedded for this build: **SongtiSC**. Technical identifiers (API paths, IDs, code) remain in English by design. | 本版嵌入中文字体：SongtiSC。技术标识符（API 路径、ID、代码）按惯例保留英文。 |

# 1. Contents

## 1. 目录

# 2. Product Intent and Design Principles

## 2. 产品定位与设计原则

Quant Research OS is an autonomous AI quantitative research laboratory. It is not a generic chat demo and not a retail trading app. The primary user is a professional researcher who must answer: what is being researched, why, which experiments ran, what survived adversarial review, how candidates interact with the existing book, and what risks remain.

Quant Research OS 是一套自主式 AI 量化研究实验室。它不是通用聊天演示，也不是零售交易 App。主要用户是专业研究员，需要回答：系统在研究什么、为什么研究、跑了哪些实验、哪些想法通过了对抗性审查、候选策略与现有组合如何相互作用、还剩哪些风险。

### 2.1 Design principles

### 2.1 设计原则

- **Precision & trust** — numbers come from engines with provenance IDs.
- **Transparency** — every transition is traced; reports link to experiments.
- **Controllability** — budgets, cooperative cancel, explicit state machine.
- **Information density** — workstation UI optimized for hours of daily use.
- **No metric invention** — agents may narrate; they must not invent Sharpe.

精确与信任 — 数字来自带溯源 ID 的引擎。

透明 — 每次状态转换可追踪；报告链接到实验。

可控 — 预算、协作式取消、显式状态机。

信息密度 — 工作站界面为每日长时间使用优化。

禁止编造指标 — 智能体可以叙述，但不得编造夏普比率。

### 2.2 Architectural hard boundary

### 2.2 架构硬边界

Language models are strong at planning, critique, and synthesis, but unreliable as calculators for portfolio mathematics. Therefore the system splits responsibilities.

语言模型擅长规划、批评与综合，但作为组合数学计算器并不可靠。因此系统明确拆分职责。

```
LLM / planner / orchestrator  →  proposes hypotheses, chooses tools, writes narrative
Deterministic quant engine     →  backtests, metrics, risk, correlations, gates
# [redacted]/[redacted] → [redacted]
# [redacted] → [redacted]
```

# 3. Repository Map

## 3. 仓库结构

| Path [redacted]<br>Path | Role [redacted]<br>Role |
|---|---|
| src/.../domain/<br>领域模型目录 | Pydantic domain objects<br>Pydantic 领域对象 |
| src/.../engine/<br>引擎目录 | Backtest, metrics, WF, robustness…<br>回测、指标、滚动验证、稳健性等 |
| src/.../agents/<br>智能体目录 | Deterministic planner/reviewer<br>确定性规划/审查（+未来 LLM 提示词） |
| src/.../tools/<br>工具目录 | Allowlisted tool router<br>白名单工具路由 |
| src/.../orchestration/<br>编排目录 | State machine + budgets<br>状态机 + 预算 |
| src/.../alpha/<br>Alpha 目录 | Alpha / strategy registry<br>Alpha / 策略注册表 |
| src/.../experiments/<br>实验目录 | Experiment registry + hashing<br>实验注册与配置哈希 |
| src/.../storage/<br>存储目录 | SQLite, paths, traces<br>SQLite、路径、轨迹 |
| src/.../api/<br>API 目录 | FastAPI<br>FastAPI 接口 |
| web/<br>前端目录 | Next.js workstation UI<br>Next.js 研究工作站 |

| | |
|---|---|
| tests/<br>测试目录 | Invariant + integration tests<br>不变量与集成测试 |
| docs/<br>文档目录 | Architecture, audit, this handbook<br>架构、审计与本手册 |

## 4. End-to-End Architecture
## 4. 端到端架构

A research question enters through CLI, API, or the workstation command bar. The orchestrator creates a ResearchRequest, plans hypotheses, validates data, runs budgeted experiments through the tool layer, validates statistically, stresses robustness and regimes, checks diversification versus existing alphas, runs adversarial review, evaluates portfolio/risk impact, and emits a report with traces.

研究问题可通过 CLI、API 或工作站命令栏进入。编排器创建 ResearchRequest，规划假设，校验数据，经工具层在预算内运行实验，做统计验证、稳健性与市场状态压力测试，检查相对既有 Alpha 的分散性，进行对抗性审查，评估组合/风险影响，并输出带轨迹的报告。

System Architecture / 系统架构

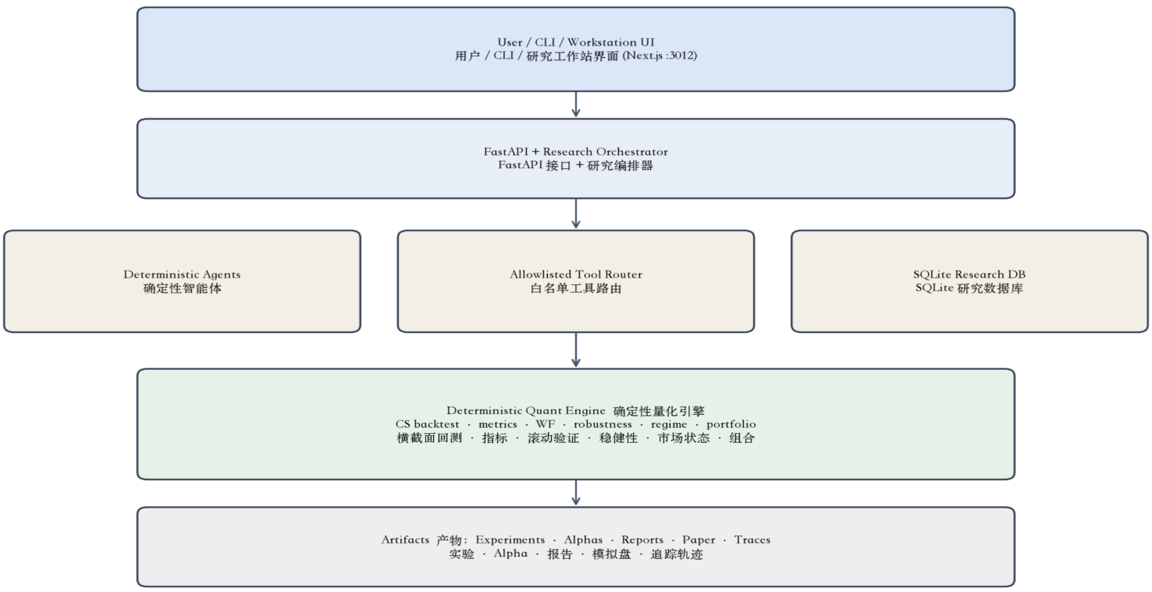


Figure 1. Layered system architecture.
图 1. 分层系统架构。

## 5. Domain Model
## 5. 领域模型

Domain objects are typed Pydantic models. They form the contract between storage, engines, API, and UI.

- **ResearchRequest** — user question, universe, budgets, status lifecycle.
- **ResearchPlan** — interpreted objectives, constraints, hypotheses.
- **Experiment** — strategy + parameters + dataset + status + metric linkage.
- **BacktestResult** — equity/returns artifact + performance metrics.
- **Strategy / Alpha** — reusable definition vs promoted research candidate.
- **ReviewResult** — adversarial findings with severity.
- **ResearchReport** — executive summary + sections + survivors/rejects.

领域对象是带类型的 Pydantic 模型，构成存储、引擎、API 与 UI 之间的契约。

ResearchRequest — 用户问题、标的宇宙、预算、状态生命周期。

ResearchPlan — 系统解释后的目标、约束与假设。

Experiment — 策略 + 参数 + 数据集 + 状态 + 指标关联。

BacktestResult — 权益/收益产物 + 绩效指标。

Strategy / Alpha — 可复用定义 vs 晋升后的研究候选。

ReviewResult — 带严重级别的对抗性发现。

ResearchReport — 执行摘要 + 章节 + 幸存者/否决项。

### 5.1 Alpha lifecycle
### 5.1 Alpha 生命周期

```
CANDIDATE → PROMISING → ROBUST → PAPER_TRADING → (LIVE future)
                          ■ REJECTED / RETIRED
# ■■ → ■■■ → ■■ → ■■■ →■■■■■■
#           ■ ■■ / ■■
```

Promotion requires **metrics_source_ids** so every Sharpe/drawdown claim is traceable to a concrete backtest/experiment artifact. This is an anti-hallucination control.

晋升要求提供 metrics_source_ids，使每条夏普/回撤声明都能追溯到具体回测/实验产物。这是反幻觉（防编造数字）控制。

## 6. Research Orchestrator

## 6. 研究编排器

The orchestrator is an explicit finite state machine (not an unbounded agent loop). Transitions are logged to the trace table. Budgets limit hypotheses/experiments. Cancel is cooperative: status becomes CANCELLED and is checked between hypotheses.

编排器是显式有限状态机（不是无界智能体循环）。转换写入轨迹表。预算限制假设/实验数量。取消是协作式的：状态设为 CANCELLED，并在假设之间检查。

Research Orchestrator State Machine / 研究编排状态机

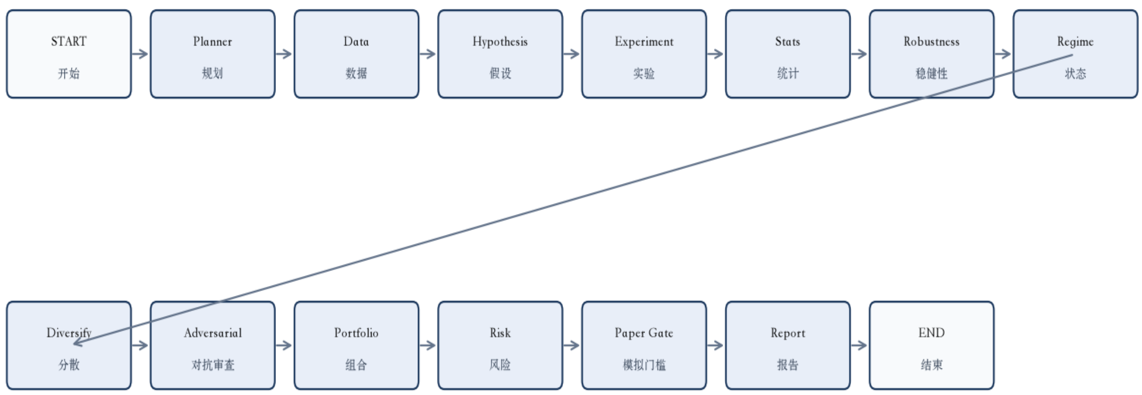


Figure 2. Orchestrator nodes from START to END.
图 2. 编排器节点：从 START 到 END。

### 6.1 Why a state machine?

### 6.1 为什么用状态机？

Autonomous research systems fail when agents recurse forever, skip validation, or mutate goals silently. An explicit transition table makes the laboratory auditable.

当智能体无限递归、跳过验证或静默改变目标时，自主研究系统会失败。显式转换表使实验室可审计。

```
TRANSITIONS = [
  START, ResearchPlanner, DataDiscovery, HypothesisGeneration,
  ExperimentDesign, ExperimentExecution, StatisticalValidation,
  RobustnessTesting, RegimeAnalysis, DiversificationAnalysis,
  AdversarialReview, PortfolioAnalysis, RiskReview,
  PaperTradingGate, Report, END,
]
```

## 7. Agents and Tool Allowlist

## 7. 智能体与工具白名单

Current production mode uses **deterministic agents** (`agents/core.py`). `agents/prompts.py` reserves future LLM mode. The UI System Health panel labels this honestly as “LLM services: deterministic agents”.

当前生产模式使用确定性智能体（`agents/core.py`）。`agents/prompts.py` 预留给未来 LLM 模式。UI 系统健康面板如实标注为“LLM 服务：确定性智能体”。

## 7.1 Planner

### 7.1 规划器

`plan_research()` expands a question into a fixed hypothesis set for the flagship FX workflow (momentum, reversal, carry-like structures, low-correlation constraints). It is a template, not an LLM call.

`plan_research()` 将问题展开为旗舰外汇流程的固定假设集合（动量、反转、类套利结构、低相关约束）。它是模板，不是 LLM 调用。

## 7.2 Tool router

### 7.2 工具路由

All quantitative side effects go through ToolRouter with an allowlist. Agents cannot call arbitrary Python. Typical tools: list/inspect/validate datasets, run_backtest, walk-forward, robustness batteries, correlation versus existing library, stress tests.

所有量化副作用都经过带白名单的 ToolRouter。智能体不能调用任意 Python。典型工具：列出/检查/校验数据集、run_backtest、滚动验证、稳健性套件、相对既有库的相关性、压力测试。

# 8. Cross-Sectional Engine & Anti Look-Ahead
# 8. 横截面引擎与反前视

Cross-sectional strategies rank a universe at each rebalance and hold a long/short basket. In FX G10, a classic example is momentum: buy recent winners, sell losers. The scientific risk is **look-ahead bias** — using information not available at decision time.

横截面策略在每次再平衡时对一篮子标的排序，并持有多空组合。在 FX G10 中，经典例子是动量：买入近期赢家、卖出输家。科学风险是前视偏差（look-ahead bias）——使用决策时尚不可得的信息。

## 8.1 Timing contract
## 8.1 时序契约

- At bar *t* close, compute signal from prices available at *t*.
- Schedule target weights for execution at *t* + *execution_lag*.
- Default execution_lag ≥ 1 → no same-bar return on new weights.
- Between rebalances, share holdings are fixed; weights drift with prices.
- Turnover is measured in weight space after NAV renormalization.

在 K 线 t 收盘，仅用 t 时刻可得价格计算信号。

将目标权重安排在 t + execution_lag 执行。

默认 execution_lag ⩾ 1 → 新权重不得计入同一根 K 线收益。

再平衡之间固定持股数量；权重随价格漂移。

换手率在净值再归一化后的权重空间中计量。

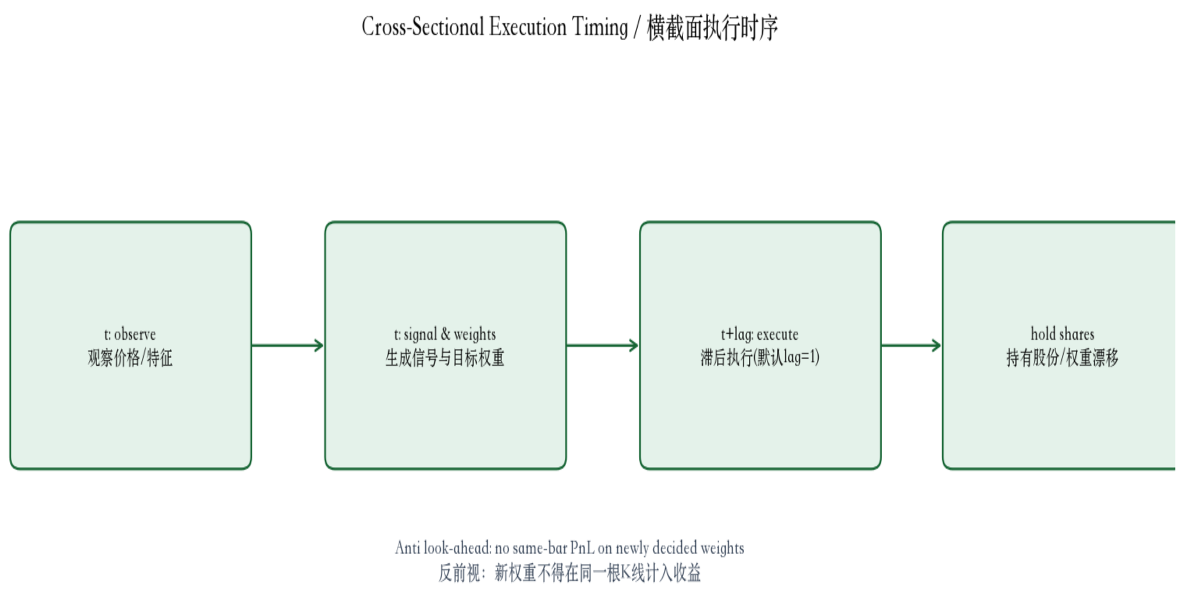


Figure 3. Signal → lag → execution → drift.
图 3. 信号 → 滞后 → 执行 → 漂移。

## 8.2 Selection methods
## 8.2 选股/选汇方法

Supported constructions include top/bottom-N, percentile baskets, z-score weights, and rank weights. Signals include momentum and short-term reversal, plus feature generators from `features/library.py` (carry, volatility rank, value proxy, liquidity stability).

支持 top/bottom-N、分位数篮子、z-score 加权与秩加权。信号包括动量与短期反转，以及 `features/library.py` 特征生成器（carry、波动排序、价值代理、流动性稳定）。

## 8.3 Weight-space turnover & NAV renormalize
## 8.3 权重空间换手与净值再归一

- Target weights are decided on rebalance bars, then shifted by execution_lag.
- One-way turnover = 0.5 · Σ|w_new – w_old| (weight space, not share counts).
- Gross portfolio return uses beginning-of-bar holdings · asset returns.

目标权重在再平衡日决定，再按 execution_lag 平移。

单边换手 = 0.5 · Σ|w_new　w_old|（权重空间，非股数）。

组合毛收益 = 期初持仓 · 资产收益。

- Holdings drift with asset returns, then divide by (1 + port_ret) so next weights stay NAV-relative.
- Net returns = portfolio returns – cost series from turnover × cost rate.

持仓随资产收益漂移，再除以 (1 + port_ret)，使次日权重相对净值。

净收益 = 组合收益 由换手 × 成本率得到的成本序列。

> If execution_lag were allowed to be 0, same-bar PnL on new weights would silently invent edge. The engine raises if lag < 1.
>
> 若允许 execution_lag=0，新权重同日盈亏会静默伪造优势。引擎在 lag < 1 时抛错。

# 9. Costs, Metrics, Validation Engines
# 9. 成本、指标与验证引擎

## 9.1 Transaction costs
## 9.1 交易成本

Costs are first-class in `engine/costs.py`. Three FX presets — OPTIMISTIC (0.85 bps), BASELINE (2.75 bps), PESSIMISTIC (10 bps) variable — multiply one-way turnover. Strategies must remain attractive under BASELINE; promotion narratives require survival under PESSIMISTIC stress. See deep dive §20 for the full economics table.

成本在 `engine/costs.py` 中是一等公民。三种外汇预设 — OPTIMISTIC（0.85 bps）、BASELINE（2.75 bps）、PESSIMISTIC（10 bps）可变 —乘以单边换手。策略在 BASELINE 下仍需有吸引力；晋升叙述需经受 PESSIMISTIC 压力。完整经济学表见深入 § 20。

## 9.2 Performance metrics
## 9.2 绩效指标

Metrics are computed exclusively in `engine/metrics.py` with pinned definitions (sibling libraries disabled — `sibling_metrics_available()` → `False`). Critical pin: population std (`ddof=0`). Sortino uses full-sample downside deviation √mean(min(r,0)²), not std of negatives only. Deep dive §18 lists the exact formulas and code contract.

指标仅在 `engine/metrics.py` 按固定定义计算（兄弟库禁用 — `sibling_metrics_available()` → `False`）。关键固定：总体标准差（`ddof=0`）。索提诺使用全样本下行偏差 √mean(min(r,0) )，而非仅负收益子集标准差。深入 § 18 列出精确公式与代码契约。

- **Annual return / volatility** — 252-day annualization.
- **Sharpe** — mean excess / std(ddof=0) · √252.
- **Sortino** — full-sample downside deviation.
- **Max drawdown / Calmar** — path-dependent risk.
- **Turnover / costs / trade_count** — implementability.

年化收益 / 波动率 — 252 日年化。

夏普 — 超额均值 / std(ddof=0) · √252。

索提诺 — 全样本下行偏差。

最大回撤 / 卡玛 — 路径依赖风险。

换手 / 成本 / 成交笔数 — 可实施性。

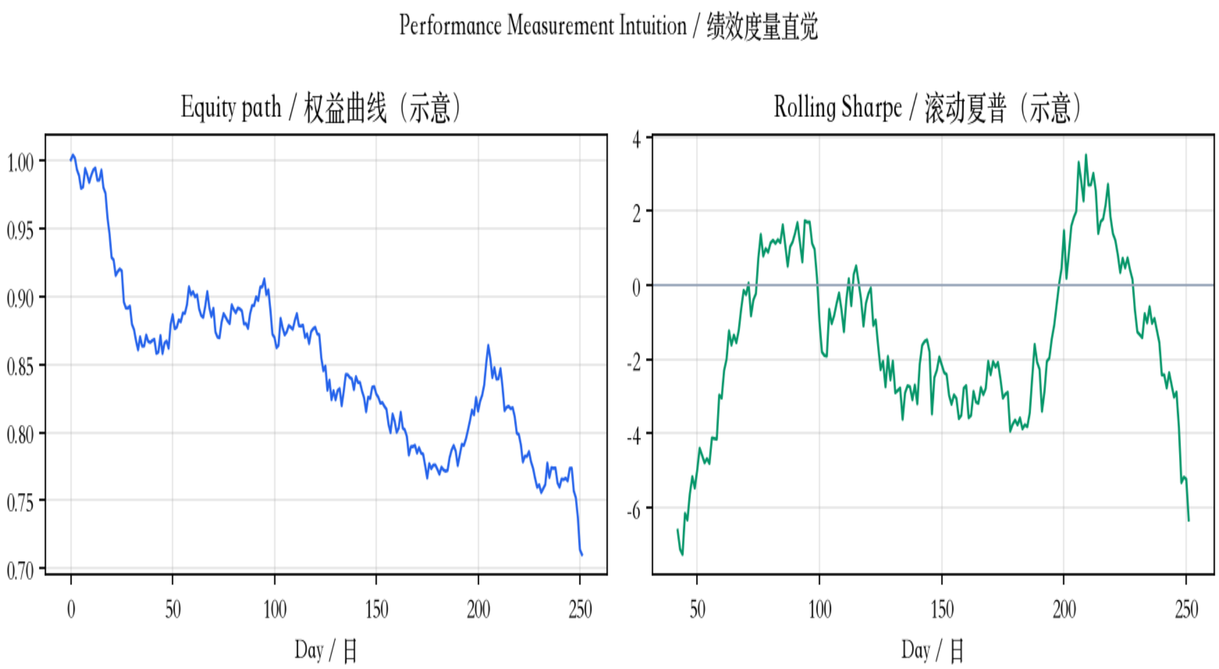


Figure 4. Intuition for equity path and rolling Sharpe.
图 4. 权益路径与滚动夏普的直觉示意。

### 9.3 Walk-forward

### 9.3 滚动前瞻验证（Walk-forward）

Walk-forward (`engine/walk_forward.py`) evaluates a **frozen** cross-sectional rule on expanding or rolling windows (defaults: 126 train / 63 test / step 63). Parameters are not re-optimized inside train windows. OOS segments are de-duplicated before aggregate Sharpe. OOS — not in-sample — is the primary promotion signal. Deep dive §27.

Walk-forward（`engine/walk_forward.py`）在扩展或滚动窗上评估冻结横截面规则（默认：126 训练 / 63 测试 / 步长 63）。训练窗内不重新优化参数。汇总夏普前对样本外片段去重。晋升看 OOS 而非样本内。深入 § 27。

### 9.4 Robustness

### 9.4 稳健性

Robustness (`engine/robustness.py`) sweeps lookback (and similar) grids and flags **fragile** knife-edge peaks (interior peak Sharpe – neighbor mean > 0.5 with peak > 0.3) and jagged surfaces (mean |ΔSharpe| ≥ 0.35). Adversarial review treats fragility as a promotion blocker. Deep dive §28.

稳健性（`engine/robustness.py`）扫描 lookback（等）网格，标记脆弱刀锋峰（内部峰值夏普 邻域均值 > 0.5 且峰值 > 0.3）与锯齿曲面（平均 |Δ夏普| ≥ 0.35）。对抗性审查将脆弱性视为晋升阻断。深入 § 28。

### 9.5 Regimes

### 9.5 市场状态（Regime）

Regime labels (`engine/regime.py`) combine rolling-vol tertiles with risk-on/off trend, then shift by 1 bar to avoid same-day attribution leakage. Labels are heuristic (confidence ~0.4). Concentration across a single regime slice is flagged. Deep dive §29.

市场状态标签（`engine/regime.py`）结合滚动波动三分位与风险偏好趋势，再平移 1 根 K 线以避免同日归因泄漏。标签为启发式（置信度 ~0.4）。绩效集中在单一状态会被标记。深入 § 29。

## 10. Alpha Library, Diversification, Portfolio & Risk

## 10. Alpha 库、分散化、组合与风险

The alpha library stores candidates with status, metrics, and correlation to the existing book. A seeded momentum alpha (`ALP-existing-momentum`) represents incumbent exposure for low-correlation discovery.

Alpha 库保存候选的状态、指标及与现有组合的相关性。种子动量 Alpha（`ALP-existing-momentum`）代表既有敞口，用于低相关发现。

### 10.1 Diversification analysis

### 10.1 分散化分析

Return correlations, downside correlations, and incremental Sharpe hints estimate whether a candidate is genuine diversification or a duplicate risk bet.

收益相关、下行相关与增量夏普提示，用于判断候选是真正分散，还是重复风险押注。

### 10.2 Portfolio / risk

### 10.2 组合 / 风险

The workstation Portfolio page supports what-if: compare current book vs book + Alpha X on Sharpe, correlation, and counts. Risk Center highlights concentration and high-correlation alerts.

工作站组合页支持情景分析：比较当前组合 vs 组合 + Alpha X 的夏普、相关性与数量。风险中心突出集中度与高相关告警。

## 11. Paper Trading Honesty Model

## 11. 模拟盘诚实模型

Paper trading in v0.2 may simulate paths using IID noise calibrated to backtest μ/σ. This is explicitly labeled in payloads and UI banners. It is useful for monitoring plumbing, but it is **not** a claim of realistic market replay.

v0.2 的模拟盘可能使用按回测 μ / σ 校准的 IID 噪声路径。这在载荷与 UI 横幅中明确标注。它对监控链路有用，但不是真实市场回放的声明。

The UI forces a clear distinction among **BACKTEST**, **PAPER TRADING**, and **LIVE**. Confusing these modes is a primary operational failure mode in quant platforms.

UI 强制区分 回测（BACKTEST）、模拟盘（PAPER TRADING） 与 实盘（LIVE）。混淆这些模式是量化平台的主要运维失败模式之一。

# 12. Storage, Reproducibility, Lineage
## 12. 存储、可复现与血缘

SQLite (WAL mode, FK pragma, indexed JSON tables) stores research requests, plans, experiments, alphas, reports, paper state, checkpoints, and agent traces. `QROS_DATA_ROOT` controls the data directory (default `data/local`).

SQLite（WAL、外键、带索引 JSON 表）存储研究请求、计划、实验、Alpha、报告、模拟盘状态、检查点与智能体轨迹。`QROS_DATA_ROOT` 控制数据目录（默认 `data/local`）。

### 12.1 Config hashing
### 12.1 配置哈希

Experiment configs are hashed so identical parameterizations can be detected and reproduced.

实验配置会被哈希，以便发现并复现相同参数化。

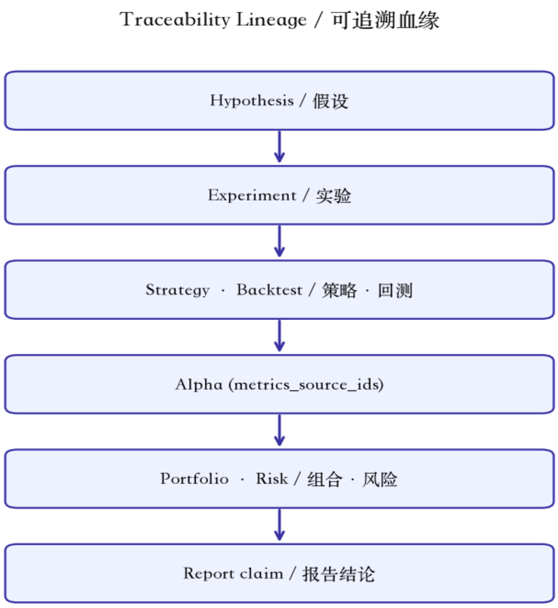


Figure 5. Claim → evidence lineage used by reports and UI.
图 5. 报告与 UI 使用的「结论 → 证据」血缘。

# 13. FastAPI Surface & Realtime
## 13. FastAPI 接口与实时

The API exposes health, research CRUD/cancel, traces, experiments, strategies, alphas, backtests, reports, portfolio, correlations, paper stepping, and SSE `/events/stream`. Optional `QROS_API_KEY`. CORS defaults include the Next.js origin :3012. Experiment list responses are enriched with metrics joined from `backtest_results`.

API 提供健康检查、研究创建/查询/取消、轨迹、实验、策略、Alpha、回测、报告、组合、相关性、模拟盘步进，以及 SSE `/events/stream`。可选 `QROS_API_KEY`。CORS 默认包含 Next.js :3012。实验列表会从 `backtest_results` 关联填充指标。

```
GET  /health
POST /research
GET  /research/{id}  ·  GET /research/{id}/trace
POST /research/{id}/cancel
GET  /experiments[?research_id=]   # includes joined metrics
GET  /alphas · /strategies · /portfolio · /paper
GET  /events/stream
```

# 14. Workstation UI Architecture
## 14. 工作站 UI 架构

The UI (`web/`) is a Next.js App Router application with a persistent shell: collapsible left navigation, top status bar, main canvas, and a bottom research command bar. Design tokens encode a dark research-terminal aesthetic (IBM Plex). Server state uses SWR.

UI（`web/`）是 Next.js App Router 应用，具有持久壳层：可折叠左侧导航、顶部状态栏、主画布与底部研究命令栏。设计令牌呈现深色研究终端美学（IBM Plex）。服务端状态使用 SWR。

Workstation Shell / 工作站壳层布局


Figure 6. Application shell composition.
图 6. 应用壳层组成。

### 14.1 Screen inventory
### 14.1 页面清单

| Route ■■<br>Route | Purpose ■■<br>Purpose |
|---|---|
| /<br>/ | Overview dashboard<br>总览仪表盘 |
| /research/[id]<br>/research/[id] | Flagship research workspace<br>旗舰研究工作区 |
| /experiments<br>/experiments | Experiment ledger + export<br>实验台账 + 导出 |
| /alphas/[id]<br>/alphas/[id] | Alpha detail + lineage<br>Alpha 详情 + 血缘 |
| /portfolio<br>/portfolio | Allocation + what-if<br>配置 + 情景分析 |
| /risk<br>/risk | Risk center<br>风险中心 |
| /agents<br>/agents | Agent observability<br>智能体可观测性 |
| /reports/[id]<br>/reports/[id] | Report reader<br>报告阅读器 |
| /memory<br>/memory | Research memory search<br>研究记忆搜索 |
| /paper<br>/paper | Paper trading (bannered)<br>模拟盘（带模式横幅） |
| Cmd+K palette<br>Cmd+K 命令面板 | Global object search<br>全局对象搜索 |

### 14.2 Frontend module boundaries

### 14.2 前端模块边界

```
web/src/app/*                 pages ■■
web/src/components/shell      chrome ■■
web/src/components/ui         design system ■■■■
web/src/components/charts     TimeSeriesChart ■■■
web/src/components/research   ResearchGraph ■■■
web/src/domain/types.ts       typed models ■■■■
web/src/lib/api.ts            API client API ■■■
web/src/lib/realtime.ts       SSE ■■
web/src/styles/*              tokens/CSS ■■■■
```

## 15. Flagship FX Workflow Walkthrough

## 15. 旗舰外汇流程走读

Flagship question: find a robust cross-sectional FX strategy with low correlation to existing momentum. The laboratory expands hypotheses, runs budgeted backtests, validates OOS, measures correlation to `ALP-existing-momentum`, adversarially reviews survivors, and writes a report.

旗舰问题：寻找与既有动量低相关、稳健的外汇横截面策略。实验室展开假设，在预算内回测，做样本外验证，测量与 `ALP-existing-momentum` 的相关性，对幸存者做对抗性审查并撰写报告。

```
quant research run \
  "Find a robust cross-sectional FX strategy with low correlation "\
  "to my existing momentum strategies." \
  --max-experiments 8

# Then open workstation Research workspace for that research_id.
# ■■■■■■■■■■ research_id ■■■■■■■
```

### 15.1 Step-by-step laboratory path

### 15.1 实验室逐步路径

- Planner emits competing FX hypotheses (carry, reversal, vol, value, liquidity) with falsifiers.
- Data agent validates synthetic FX panels and seeds ALP-existing-momentum as incumbent book.
- Each hypothesis becomes a hashed Experiment → lagged CS backtest → metrics artifact.
- Walk-forward / robustness / regime filters remove fragile winners.
- Diversification vs momentum + adversarial review assign PROMISING / REJECT / more-research.
- Report + traces enable drill-down; Portfolio what-if estimates book impact.

规划器发出竞争的外汇假设（套息、反转、波动、价值、流动性）及证伪条件。

数据智能体校验合成外汇面板，并播种 ALP-existing-momentum 作为既有组合。

每个假设变成带哈希的 Experiment → 滞后横截面回测 → 指标产物。

滚动验证 / 稳健性 / 市场状态过滤去掉脆弱赢家。

相对动量的分散分析 + 对抗性审查给出 PROMISING / REJECT / 需更多研究。

报告 + 轨迹支持下钻；组合情景分析估计账面影响。

### 15.2 How to read a completed run

### 15.2 如何阅读已完成运行

In the workstation: Overview → latest research card → Research workspace graph (all nodes completed) → Experiments tab sorted by Sharpe → open best EXP → confirm costs/lag/dataset → Agent Inspector tool payloads → Report survivors → Portfolio what-if. Never stop at the executive summary.

在工作站：总览 → 最新研究卡片 → 研究工作区图（节点均完成）→ 实验页按夏普排序 → 打开最佳 EXP → 确认成本/滞后/数据集 → 智能体检查器工具载荷 → 报告幸存者 → 组合情景分析。切勿只看执行摘要。

## 16. Testing, Operations, Known Gaps

## 16. 测试、运维与已知缺口

### 16.1 Verification snapshot

### 16.1 验证快照

- Backend: pytest — 32 passed.
- Frontend: vitest + next build — passed.
- Runtime smoke: API healthy; UI routes HTTP 200; browser confirmed Overview, Research graph, Experiments with Sharpes, Alpha Library, Portfolio.

后端：pytest — 32 通过。

前端：vitest + next build — 通过。

运行冒烟：API 健康；UI 路由 HTTP 200；浏览器确认总览、研究图、带夏普的实验表、Alpha 库、组合页。

• Fixes during verification: normalize `user_question`; enrich experiment metrics from backtests.

验证中修复：规范化 `user_question`；从回测关联填充实验指标。

## 16.2 Runbook
## 16.2 运行手册

```
# API
source .venv/bin/activate
quant serve --host 127.0.0.1 --port 8002

# UI
cd web && npm install && npm run dev   # http://127.0.0.1:3012

# Handbook PDF
python docs/generate_handbook_pdf.py
```

## 16.3 Known gaps (honest)
## 16.3 已知缺口（如实）

• Planner is a deterministic template — not a live LLM.

• Paper trading may use IID noise simulation (labeled).

• Full resume-from-checkpoint node restart is incomplete.

• True async worker fleet / RBAC / Docker hardening incomplete.

• Some UI charts are metric-conditioned until full equity series attach.

规划器是确定性模板——不是实时 LLM。

模拟盘可能使用 IID 噪声仿真（已标注）。

从检查点按节点完整恢复尚未完成。

真正的异步工作集群 / RBAC / Docker 加固未完成。

部分 UI 图表在完整权益序列接入前由指标条件生成示意。

# 17. Appendix — Key Code References
# 17. 附录 — 关键代码索引

```
orchestration/runner.py      state machine + budgets + cancel  ■■■/■■/■■
agents/core.py               plan_research, adversarial_review ■■■■■■■■
tools/router.py              allowlisted tools ■■■■■
engine/cross_sectional.py    lagged CS backtest ■■■■■■■■
engine/metrics.py            pinned performance math ■■■■■■
engine/walk_forward.py       OOS evaluation ■■■■■
engine/robustness.py         sensitivity batteries ■■■■■
engine/regime.py             shifted regime labels ■■■■■■■■
alpha/registry.py            alpha/strategy lifecycle ■■■■
experiments/registry.py      experiment persistence ■■■■■
storage/db.py                SQLite + traces + checkpoints
api/app.py                   HTTP + SSE + enrichment
web/src/app/research/[id]    flagship workspace UI ■■■■■
web/src/lib/api.ts           typed client ■■■■■■
```

# 18. Deep Dive: Metrics Mathematics

## 18. 深入解析：指标数学

All performance numbers in Quant Research OS are computed exclusively in `engine/metrics.py`. Sibling metric libraries are **disabled** (`sibling_metrics_available() → False`) because different `ddof` conventions silently change Sharpe. The pinned formulas below are the contract — agents and tools must not invent numbers.

Quant Research OS 中的全部绩效数字仅在 `engine/metrics.py` 计算。兄弟指标库被禁用（`sibling_metrics_available() → False`），因为不同的 `ddof` 约定会静默改变夏普。以下固定公式即为契约——智能体与工具不得编造数字。

### 18.1 Equity path and annualization

### 18.1 权益路径与年化

Given net daily returns `r_t`, equity is `E_t = Π(1 + r_t)`. Cumulative return is `E_T − 1`. Annualized return uses `(1 + cum)^(252/N) − 1` with N = number of daily observations and 252 trading days per year.

给定净日收益 `r_t`，权益为 `E_t = Π(1 + r_t)`。累计收益为 `E_T − 1`。年化收益使用 `(1 + cum)^(252/N) − 1`，N 为日观测数，每年 252 个交易日。

### 18.2 Sharpe, Sortino, MDD (pinned)

### 18.2 夏普、索提诺、最大回撤（固定）

- **Volatility**: `σ = std(r, ddof=0) · √252`. Population standard deviation (ddof=0) is pinned — not sample std.
- **Sharpe**: `mean(r − r_f/252) / std(r, ddof=0) · √252`. Excess return over daily risk-free; denominator uses full-sample std with ddof=0.
- **Sortino**: downside deviation = `√mean(min(r, 0)²)` over the **full sample** (zero returns contribute). Not std of negative returns only.
- **Max drawdown (MDD)**: `min(E_t / cummax(E) − 1)`.
- **Calmar**: annualized return / |MDD| when MDD < 0.

- 波动率：`σ = std(r, ddof=0) · √252`。固定使用总体标准差（ddof=0）——非样本标准差。
- 夏普：`mean(r − r_f/252) / std(r, ddof=0) · √252`。超额为日无风险利率之上；分母为全样本 ddof=0 标准差。
- 索提诺：下行偏差 = 全样本上的 `√mean(min(r, 0)²)`（零收益也计入）。不是仅对负收益子集求标准差。
- 最大回撤（MDD）：`min(E_t / cummax(E) − 1)`。
- 卡玛：当 MDD < 0 时，年化收益 / |MDD|。

```
# engine/metrics.py — pinned implementation
equity = (1 + r).cumprod()
ann_ret = (1 + cum) ** (252 / len(r)) - 1
vol = r.std(ddof=0) * sqrt(252)
excess = r - risk_free_rate / 252
sharpe = excess.mean() / r.std(ddof=0) * sqrt(252)
downside = np.minimum(r, 0.0)
down_dev = sqrt(np.mean(downside ** 2))   # full-sample
sortino = excess.mean() / down_dev * sqrt(252)
mdd = (equity / equity.cummax() - 1.0).min()
calmar = ann_ret / abs(mdd) if mdd < 0 else 0.0
```

> If a reported Sharpe cannot be reproduced by these formulas on the linked return series, the number is invalid — regardless of narrative quality.
>
> 若报告的夏普无法用这些公式在关联收益序列上复现，则该数字无效——无论叙述多么有说服力。

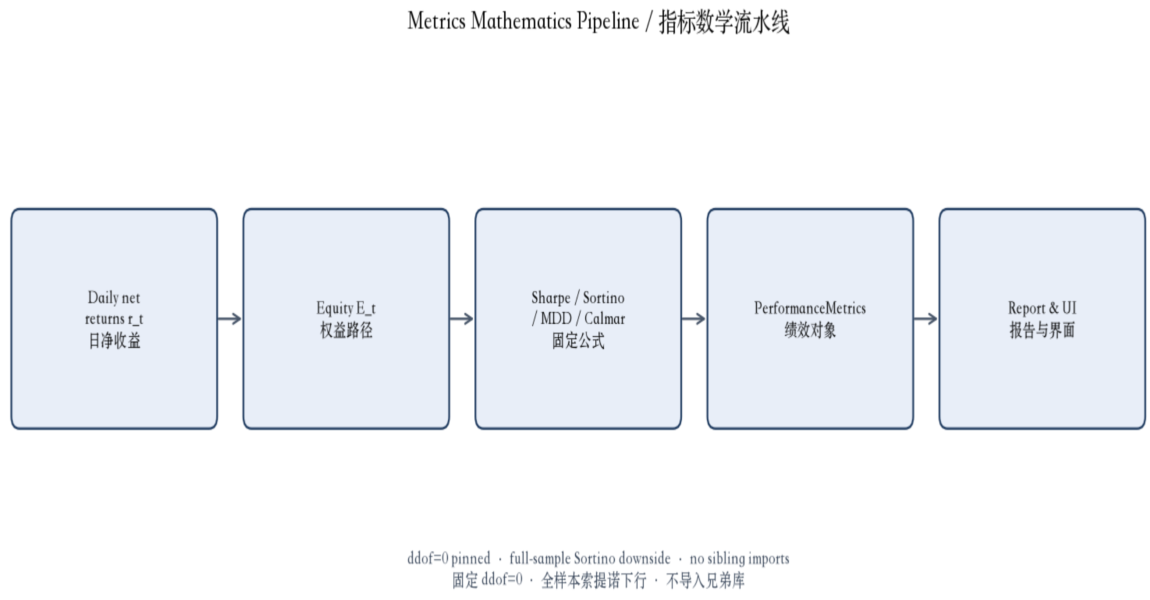


Figure 7. Pinned metrics pipeline from net returns to report fields.
图 7. 从净收益到报告字段的固定指标流水线。

## 19. Deep Dive: Backtest Loop Internals
## 19. 深入解析：回测循环内部

File: `engine/cross_sectional.py`. The cross-sectional engine constructs long/short baskets from scores and simulates portfolio returns under a strict anti-look-ahead timing contract: signal at bar t, execution at `t + execution_lag` (≥ 1), weight-space turnover, NAV renormalization, and transaction costs.

文件：`engine/cross_sectional.py`。横截面引擎由分数构建多空篮子，并在严格反前视时序契约下模拟组合收益：t 时刻信号、t + `execution_lag`（≥ 1）执行、权重空间换手、净值再归一化与交易成本。

### 19.1 Configuration surface
### 19.1 配置面

```
class CrossSectionalConfig(BaseModel):
    lookback: int = 20
    top_n: int = 3
    bottom_n: int = 3
    selection: TOP_BOTTOM_N | PERCENTILE | ZSCORE_WEIGHT | RANK_WEIGHT
    rebalance_every: int = 5
    execution_lag: int = Field(default=1, ge=1)  # hard floor
    gross_exposure: float = 1.0
    cost_assumption: optimistic | baseline | pessimistic
    signal_name: str = "momentum"  # or reversal / feature library
```

### 19.2 Selection methods
### 19.2 选择方法

| Method<br>方法 | Mechanism<br>机制 | When to use<br>适用场景 |
|---|---|---|
| TOP_BOTTOM_N<br>Top/Bottom N | Long top N, short bottom N with equal split<br>多头 top N、空头 bottom N 等权 | Default FX basket; interpretable book<br>默认外汇篮子；可解释持仓 |
| PERCENTILE<br>分位数 | Long above percentile_long; short below percentile_short<br>高于上分位做多；低于下分位做空 | Smooth basket sizing across universe<br>跨宇宙平滑篮子规模 |
| ZSCORE_WEIGHT<br>Z 分数加权 | Center scores; weight proportional to z-score<br>中心化分数；按 z 分数比例加权 | Continuous exposure to signal strength<br>对信号强度连续暴露 |
| RANK_WEIGHT<br>秩加权 | Center ranks; weight proportional to rank<br>中心化秩；按秩比例加权 | Robust to outliers in raw scores<br>对原始分数异常值稳健 |

### 19.3 Simulation loop (critical)
### 19.3 仿真循环（关键）

- On rebalance bars, compute target weights from cross-sectional scores.
- Shift targets by `execution_lag` bars (≥ 1). Default lag=1 → no same-bar PnL on newly decided weights.
- Turnover in **weight space**: `turnover = 0.5 · Σ|w_new − w_old|`. One-way traded fraction.
- Portfolio return: `port_ret = Σ(w · r_asset)` using beginning-of-bar weights.
- **NAV renormalize**: drift holdings by asset returns, then divide by `(1 + port_ret)` so next-day weights remain return-relative.
- Subtract cost series derived from turnover × cost rate.

在再平衡日由横截面分数计算目标权重。

将目标按 `execution_lag` 根 K 线平移（≥ 1）。默认 lag=1 → 新权重不得在同一根 K 线计入收益。

权重空间换手：`turnover = 0.5 · Σ|w_new − w_old|`。单边交易比例。

组合收益：`port_ret = Σ(w · r_asset)`，使用期初权重。

净值再归一：按资产收益漂移持仓，再除以 `(1 + port_ret)`，使次日权重仍相对净值。

减去由换手 × 成本率导出的成本序列。

```
# Anti look-ahead guard
if cfg.execution_lag < 1:
    raise ValueError("execution_lag must be >= 1")
scheduled = target.shift(cfg.execution_lag)
turnover = abs(new_w - cur_w).sum() / 2.0   # weight space
port_ret = (holdings * asset_rets).sum()
holdings = holdings * (1 + asset_rets) / (1 + port_ret)  # NAV renormalize
net = ret_series - cost_series
```

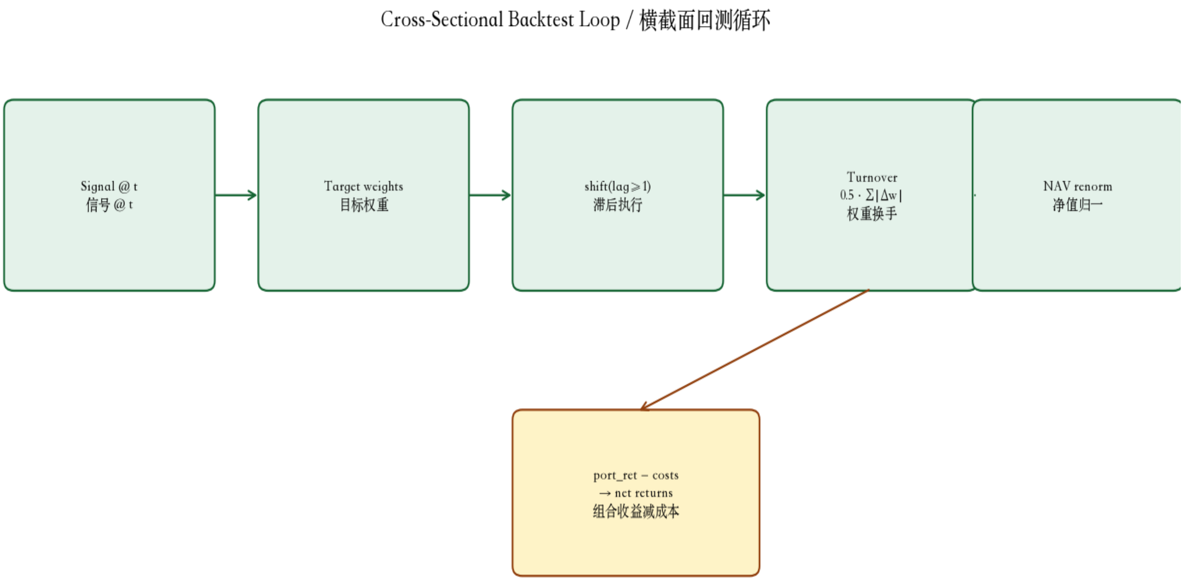


Figure 8. One bar of the cross-sectional simulation loop.
图 8. 横截面仿真循环中的一根 K 线。

## 20. Deep Dive: Cost Model Economics
## 20. 深入解析：成本模型经济学

File: `engine/costs.py`. Transaction costs are applied to one-way turnover (`0.5 · Σ|Δw|`). Three FX-oriented presets bracket realistic execution: OPTIMISTIC, BASELINE, and PESSIMISTIC. Variable cost rate = (proportional + spread + slippage) bps / 10,000.

文件：`engine/costs.py`。交易成本作用于单边换手（`0.5 · Σ|Δw|`）。三种外汇取向预设框定现实执行：OPTIMISTIC（乐观）、BASELINE（基准）、PESSIMISTIC（悲观）。可变成本率 = (比例 + 点差 + 滑点) bps / 10,000。

| Assumption<br>假设 | Prop bps<br>比例 bps | Spread bps<br>点差 bps | Slippage bps<br>滑点 bps | Total var bps<br>可变合计 bps |
|---|---|---|---|---|
| OPTIMISTIC<br>乐观 | 0.5<br>0.5 | 0.25<br>0.25 | 0.1<br>0.1 | 0.85<br>0.85 |

| | | | | |
|---|---|---|---|---|
| BASELINE<br>基准 | 1.5<br>1.5 | 0.75<br>0.75 | 0.5<br>0.5 | 2.75<br>2.75 |
| PESSIMISTIC<br>悲观 | 5.0<br>5.0 | 3.0<br>3.0 | 2.0<br>2.0 | 10.0<br>10.0 |

Cost per rebalance ≈ `turnover × variable_bps / 10,000`. Strategies must remain attractive under BASELINE costs; promotion pressure requires survival under PESSIMISTIC assumptions. Walk-forward and robustness batteries re-run with cost stress.

每次再平衡成本 ≈ `turnover × variable_bps / 10,000`。策略在 BASELINE 成本下仍需有吸引力；晋升压力要求经受 PESSIMISTIC 假设。滚动验证与稳健性套件以成本压力重跑。

> A strategy with Sharpe 2.0 under OPTIMISTIC but negative Sharpe under PESSIMISTIC is a cost-sensitivity failure, not a candidate for promotion.
>
> 在 OPTIMISTIC 下夏普 2.0、在 PESSIMISTIC 下夏普为负的策略是成本敏感失败，不是晋升候选。

# 21. Deep Dive: Orchestrator Experiment Loop

## 21. 深入解析：编排器实验循环

File: orchestration/runner.py. After plan_research() and dataset validation, the runner iterates candidate hypotheses. Each iteration is budget-gated (budget.can_run_experiment()) and cancel-aware (db.is_cancelled(research_id) checked between hypotheses).

文件：orchestration/runner.py。在 plan_research() 与数据集校验之后，运行器遍历候选假设。每次迭代受预算门控（budget.can_run_experiment()）并感知取消（假设间检查 db.is_cancelled(research_id)）。

### 21.1 Feature → dataset mapping

### 21.1 特征 → 数据集映射

| Feature<br>特征 | SIGNAL_MAP key<br>映射键 | Dataset<br>数据集 | Lookback (typical)<br>回看（典型） |
|---|---|---|---|
| carry<br>carry | cross_sectional_carry<br>cross_sectional_carry | fx_synthetic_momentum<br>fx_synthetic_momentum | 60<br>60 |
| reversal<br>reversal | short_term_reversal<br>short_term_reversal | fx_synthetic_meanrev<br>fx_synthetic_meanrev | 1<br>1 |
| volatility<br>volatility | volatility_ranked<br>volatility_ranked | fx_synthetic_momentum<br>fx_synthetic_momentum | 20<br>20 |
| value<br>value | value_proxy<br>value_proxy | fx_synthetic_meanrev<br>fx_synthetic_meanrev | 20<br>20 |
| liquidity<br>liquidity | liquidity_stability<br>liquidity_stability | fx_synthetic_momentum<br>fx_synthetic_momentum | 20<br>20 |

### 21.2 Per-hypothesis pipeline

### 21.2 每个假设的流水线

- Map hypothesis name → feature via SIGNAL_MAP.
- Persist Strategy with lookback, signal, rebalance_every; link hypothesis edge.
- Create Experiment; call run_backtest via ToolRouter.
- Run walk-forward, robustness, regime, diversification vs seeded momentum alpha.
- Adversarial review emits severity-tagged findings; survivors may register as Alpha.

经 SIGNAL_MAP 将假设名映射到特征。

持久化 Strategy（含 lookback、signal、rebalance_every）；链接假设边。

创建 Experiment；经 ToolRouter 调用 run_backtest。

运行滚动验证、稳健性、市场状态、相对种子动量 Alpha 的分散性分析。

对抗性审查产出带严重级别的发现；幸存者可能注册为 Alpha。

```
# Pseudocode — orchestration/runner.py inner loop
for hyp in plan.candidate_hypotheses:
    if db.is_cancelled(research_id): break
    if not budget.can_run_experiment(): break
    feature = SIGNAL_MAP.get(hyp.name, "momentum")
    dataset = DATASET_FOR[feature]  # momentum vs meanrev panel
    strategy = Strategy(..., parameters={lookback, signal, rebalance_every})
    budget.consume_experiment()
    exp = tools.call("create_experiment", hypothesis_id=hyp.hypothesis_id, ...)
    bt  = tools.call("run_backtest", experiment_id=exp.id, dataset_id=dataset, ...)
    wf  = tools.call("walk_forward", ...)
    rob = tools.call("robustness_battery", ...)
    div = tools.call("analyze_diversification", vs="ALP-existing-momentum")
    rev = adversarial_review(candidate, evidence)
    maybe_register_alpha(bt, metrics_source_ids=[bt.backtest_id])
```

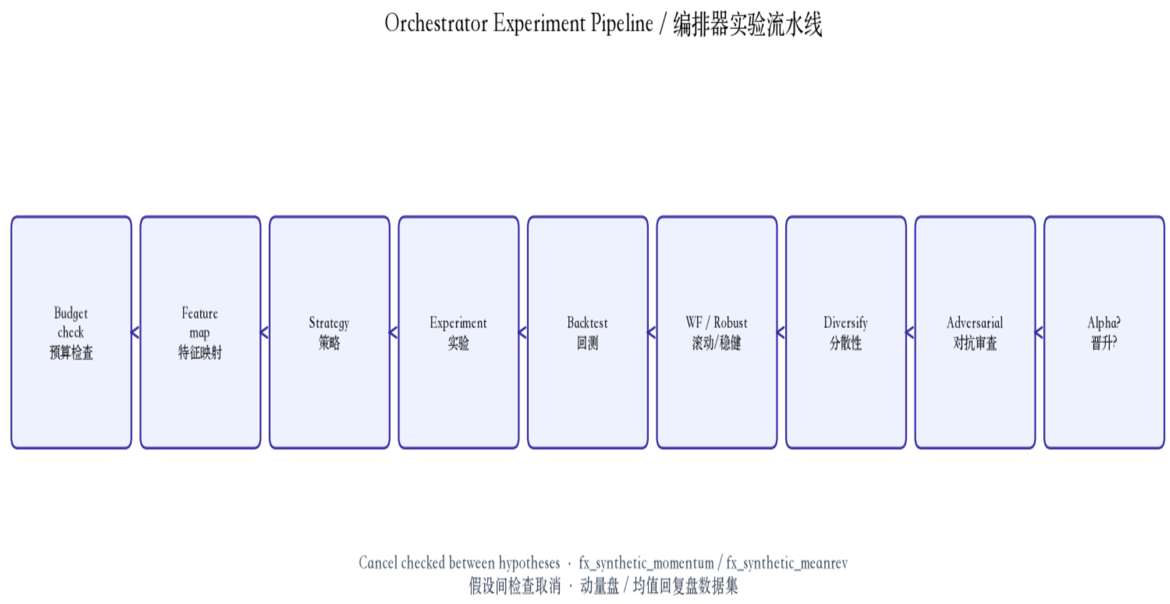


Figure 9. Budgeted hypothesis → experiment → validation loop.
图 9. 有预算的 假设→实验→验证 循环。

## 22. Deep Dive: Hypothesis Catalog & Falsification

## 22. 深入解析：假设目录与证伪

File: `agents/core.py` — `plan_research()` emits five fixed candidate hypotheses. Each carries explicit `falsification_criteria` so researchers know in advance what evidence would reject the idea.

文件：`agents/core.py` — `plan_research()` 产出五个固定候选假设。每个假设带有明确的 `falsification_criteria`，使研究员事先知道何种证据会否决该想法。

| Hypothesis<br>假设 | Economic intuition<br>经济直觉 | Falsification criteria<br>证伪标准 |
|---|---|---|
| cross_sectional_carry<br>横截面 carry | High-yield currencies earn carry premium<br>高息货币赚取 carry 溢价 | OOS Sharpe ≤ 0 after costs or corr(momentum) > 0.5<br>成本后 OOS 夏普 ≤ 0 或与动量相关 > 0.5 |
| short_term_reversal<br>短期反转 | Transitory flows reverse over short horizons<br>短期资金流反转 | Fails walk-forward or excessive turnover under costs<br>滚动验证失败或成本下换手过高 |
| volatility_ranked<br>波动率排序 | Relative vol forecasts risk-adjusted opportunity<br>相对波动预测风险调整机会 | No OOS edge; pure risk packaging without alpha<br>无 OOS 优势；纯风险包装无 alpha |
| value_proxy<br>价值代理 | Mean-reversion to longer-run fair value<br>向长期公允价值均值回复 | Collapses to slow momentum/reversal with high corr<br>坍缩为高相关的慢动量/反转 |
| liquidity_stability<br>流动性稳定 | Stable/liquid names earn premium in stress<br>稳定/高流动名称在压力下溢价 | Indistinguishable from low-vol factor; no diversification<br>与低波因子不可区分；无分散性 |

Success criteria (plan-level): OOS Sharpe > 0.3 after baseline costs, |corr(momentum)| < 0.35, parameter surface not fragile, adversarial severity < CRITICAL.

成功标准（计划级）：基准成本后 OOS 夏普 > 0.3，|corr(动量)| < 0.35，参数面不脆弱，对抗性严重度 < CRITICAL。

## 23. Deep Dive: Storage Schema & Trace Model

## 23. 深入解析：存储模式与轨迹模型

File: `storage/db.py`. SQLite stores JSON payloads with indexed columns for status queries. WAL mode and `foreign_keys=ON` are enabled. Every orchestrator action writes structured `agent_traces` rows.

文件：`storage/db.py`。SQLite 存储 JSON 载荷，索引列支持状态查询。启用 WAL 与 `foreign_keys=ON`。每个编排动作写入结构化 `agent_traces` 行。

```
research_requests(research_id PK, payload, status, created_at)
research_plans(research_id PK → requests, payload, created_at)
hypotheses(hypothesis_id PK, research_id, payload)
experiments(experiment_id PK, research_id, payload, status,
            configuration_hash, created_at)
strategies(strategy_id PK, payload, version)
alphas(alpha_id PK, strategy_id, payload, status)
backtest_results(backtest_id PK, experiment_id, payload)
validation_results(validation_id PK, experiment_id, payload)
reports(research_id PK, payload, decision, created_at)
agent_traces(id, research_id, agent, event_type, payload, created_at)
lineage_edges(src_type, src_id, rel, dst_type, dst_id, meta)
checkpoints(research_id, node, payload, ...)
```

## 23.1 Lineage edges and traces

## 23.1 血缘边与轨迹

- research —has_hypothesis→ hypothesis
- hypothesis —implements→ strategy
- strategy —used_in→ experiment
- experiment —produced→ backtest
- experiment —validated_by→ validation
- Alpha.metrics_source_ids → backtest_id(s) for provenance
- agent_traces: planner, data, orchestrator, adversarial events with JSON payloads

research —has_hypothesis→ hypothesis

hypothesis —implements→ strategy

strategy —used_in→ experiment

experiment —produced→ backtest

experiment —validated_by→ validation

Alpha.metrics_source_ids → backtest_id（们）用于溯源

agent_traces：规划、数据、编排、对抗性事件及 JSON 载荷

# 24. Deep Dive: API Contracts & UI Binding

## 24. 深入解析：API 契约与 UI 绑定

File: `api/app.py`. Base URL default: `http://127.0.0.1:8002`. Optional header `X-API-Key` when `QROS_API_KEY` is set. CORS allows workstation origin `:3012`.

文件：`api/app.py`。默认基址：`http://127.0.0.1:8002`。若设置 `QROS_API_KEY`，需头 `X-API-Key`。CORS 允许工作站来源 `:3012`。

### 24.1 Experiment enrichment

### 24.1 实验富化

`GET /experiments` returns registry rows joined with `backtest_results` metrics. Aliases `ann_return` and `ann_vol` are exposed so the workstation experiment table Sharpe column is never blank when a backtest exists.

`GET /experiments` 返回注册表行并关联 `backtest_results` 指标。暴露别名 `ann_return` 与 `ann_vol`，使工作站实验表在存在回测时夏普列不为空。

### 24.2 SWR vs UI state

### 24.2 SWR 与 UI 状态

- **Server state (SWR)**: keys like `overview-research`, `research-{id}`, `experiments-all`. Stale-while-revalidate polling.
- **UI state (React)**: selected graph node, table sort, sidebar collapse, command palette query — local component state / localStorage.
- **SSE**: `GET /events/stream` emits `research.updated`. EventSource cannot attach API keys; UI falls back to SWR when key required.

服务端状态（SWR）：键如 `overview-research`、`research-{id}`、`experiments-all`。陈旧-重验证轮询。

UI 状态（React）：选中图节点、表排序、侧栏折叠、命令面板查询 — 组件本地状态 / localStorage。

SSE：`GET /events/stream` 发出 `research.updated`。EventSource 无法附加 API key；需要密钥时 UI 回退 SWR。

```
GET /research/{id} → {
  request: { research_id, user_question, budget, status, ... },
  plan: { objectives, hypotheses, success_criteria, ... },
  report: { executive_summary, survivors, decision, ... },
  checkpoint: { node, payload } | null
}
GET /research/{id}/trace → [ { agent, event_type, payload, created_at }, ... ]
```

# 25. Deep Dive: Traceability Drill-Down Example

## 25. 深入解析：可追溯下钻示例

Worked path for a flagship FX research run (synthetic panels). Numbers vary by seed; the evidence chain structure is invariant.

旗舰外汇研究运行的走查路径（合成面板）。数字因种子而异；证据链结构不变。

| Step<br>步骤 | Entity / field<br>实体/字段 | What to inspect<br>检查内容 |
|---|---|---|
| 1<br>1 | Report.decision<br>Report.decision | REQUIRES_MORE_RESEARCH — honest outcome<br>REQUIRES_MORE_RESEARCH — 诚实结果 |
| 2<br>2 | Report.survivors[0].alpha_id<br>幸存者 alpha_id | Link to alpha registry row<br>链接 Alpha 注册表行 |
| 3<br>3 | Alpha.metrics_source_ids<br>metrics_source_ids | List of backtest_id values — provenance anchor<br>backtest_id 列表 — 溯源锚点 |
| 4<br>4 | backtest_results.payload<br>回测载荷 | Sharpe, Sortino, MDD, turnover, cost breakdown<br>夏普、索提诺、MDD、换手、成本分解 |
| 5<br>5 | experiments.configuration_hash<br>配置哈希 | Reproducibility fingerprint (params + dataset + costs)<br>可复现指纹（参数+数据+成本） |
| 6<br>6 | agent_traces (orchestrator)<br>编排轨迹 | Tool call params/results for run_backtest, walk_forward<br>run_backtest、walk_forward 工具调用参数/结果 |

- Report conclusion → survivor alpha_id
- Alpha.metrics_source_ids → backtest_id
- backtest_id → experiment_id → strategy parameters

报告结论 → 幸存者 alpha_id

Alpha.metrics_source_ids → backtest_id

backtest_id → experiment_id → 策略参数

• Trace → exact tool inputs that produced the number

轨迹 → 产生该数字的精确工具输入

Traceability Drill-Down Example / 可追溯下钻示例

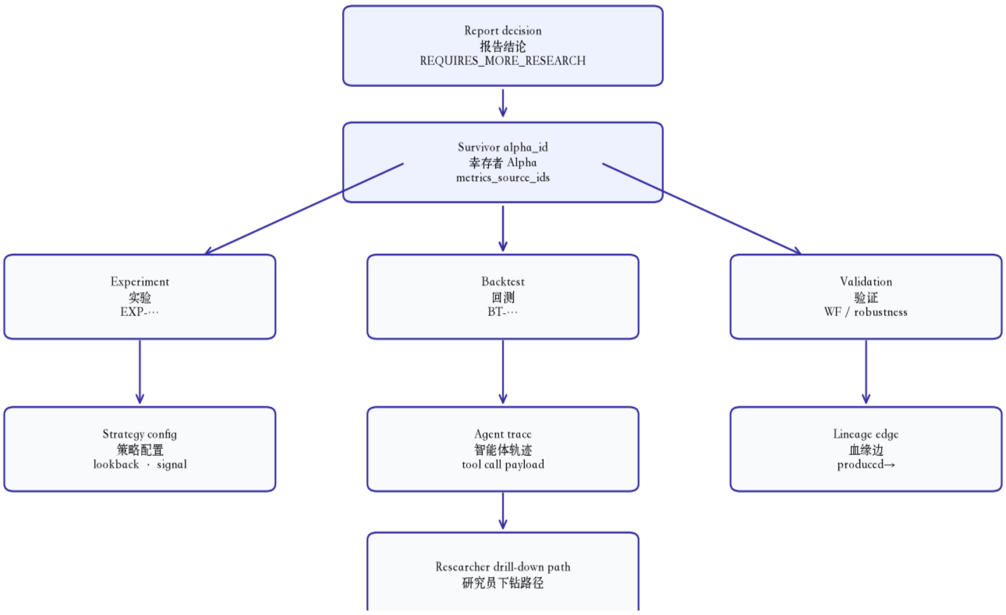


Figure 10. Report → experiment → backtest → trace drill-down.
图 10. 报告 → 实验 → 回测 → 轨迹下钻。

# 26. Threats to Validity & Research Hygiene

## 26. 效度威胁与研究卫生

Quant Research OS is a laboratory, not a live trading system. Researchers must actively guard against threats that make backtests look better than deployable reality.

**Quant Research OS 是实验室，不是实盘系统。研究员必须主动防范使回测看起来优于可部署现实的威胁。**

### 26.1 Threat catalog

### 26.1 威胁目录

| Threat<br>威胁 | Manifestation in QROS<br>在 QROS 中的表现 | Mitigation<br>缓解 |
|---|---|---|
| Look-ahead bias<br>前视偏差 | Same-bar execution on new weights<br>新权重同一根 K 线执行 | execution_lag ≥ 1 enforced in engine<br>引擎强制 execution_lag ≥ 1 |
| Overfitting / data snooping<br>过拟合/数据窥探 | Many hypotheses on same panel<br>同一面板上多假设 | Budget caps, walk-forward, adversarial review<br>预算上限、滚动验证、对抗性审查 |
| Cost underestimation<br>成本低估 | Ignoring spread/slippage<br>忽略点差/滑点 | Three-tier FX cost presets; pessimistic stress<br>三层 FX 成本预设；悲观压力 |
| Metric inconsistency<br>指标不一致 | ddof=1 Sharpe from external lib<br>外部库 ddof=1 夏普 | Pinned metrics.py; sibling imports disabled<br>固定 metrics.py；禁用兄弟库 |
| Narrative invention<br>叙述编造 | Agent prose without tool backing<br>无工具支撑的智能体 prose | Numbers only via ToolRouter; metrics_source_ids<br>数字仅经 ToolRouter；metrics_source_ids |
| Mode confusion<br>模式混淆 | PAPER / synthetic treated as LIVE<br>模拟盘/合成数据当成实盘 | UI banners; BACKTEST labels on synthetic data<br>UI 横幅；合成数据 BACKTEST 标签 |
| Survivorship in alpha library<br>Alpha 库幸存者偏差 | Only winners visible<br>仅见赢家 | Failed experiments persisted; adversarial findings surfaced<br>失败实验持久化；对抗性发现展示 |

### 26.2 Practitioner checklist

### 26.2 从业者清单

• Verify Sharpe by recomputing from stored net returns with ddof=0 formulas.

用 ddof=0 公式从已存净收益重算以验证夏普。

<table>
<tr><td>• Check OOS walk-forward Sharpe, not just in-sample.</td><td>检查 OOS 滚动验证夏普，而非仅样本内。</td></tr>
<tr><td>• Re-run under PESSIMISTIC costs before any promotion narrative.</td><td>任何晋升叙述前在 PESSIMISTIC 成本下重跑。</td></tr>
<tr><td>• Confirm |corr vs momentum| < 0.35 for diversification claims.</td><td>对分散性声明确认 |相对动量相关| < 0.35。</td></tr>
<tr><td>• Drill report → metrics_source_ids → backtest payload before trusting a number.</td><td>信任数字前先下钻 报告 → metrics_source_ids → 回测载荷。</td></tr>
<tr><td>• Read adversarial findings — HIGH/CRITICAL blocks promotion.</td><td>阅读对抗性发现 — HIGH/CRITICAL 阻断晋升。</td></tr>
<tr><td>• Treat synthetic FX panels (fx_synthetic_*) as structural demos, not live edge.</td><td>将合成 FX 面板（fx_synthetic_*）视为结构演示，非实盘优势。</td></tr>
<tr><td>• Run pytest invariants: <code>tests/test_quant_invariants.py</code>.</td><td>运行 pytest 不变量：<code>tests/test_quant_invariants.py</code>。</td></tr>
<tr><td>Extension rule: if a change lets natural language create a number that cannot be recomputed from stored returns + pinned formulas, reject the change.</td><td>扩展规则：若某改动使自然语言能产生无法由已存收益 + 固定公式重算的数字，则拒绝该改动。</td></tr>
</table>

# 27. Deep Dive: Walk-Forward Validation
# 27. 深入解析：滚动前瞻验证

File: `engine/walk_forward.py`. Walk-forward in QROS is **segmented out-of-sample evaluation of a frozen rule** — parameters in `CrossSectionalConfig` are treated as exogenous and are **not** re-fit inside each train window. This is intentionally stricter and more honest than nested parameter search that quietly overfits the train set.

文件：`engine/walk_forward.py`。QROS 中的 walk-forward 是对冻结规则的分段样本外评估——`CrossSectionalConfig` 中的参数视为外生，不会在每个训练窗内重新拟合。这比在训练集上静默过拟合的嵌套参数搜索更严格、更诚实。

## 27.1 Window modes and defaults
## 27.1 窗口模式与默认值

| Field<br>字段 | Default<br>默认 | Meaning<br>含义 |
|---|---|---|
| mode<br>mode | expanding_window<br>expanding_window | Train grows from t=0; alternate rolling_window<br>训练自 t=0 扩展；可改 rolling |
| train_bars<br>train_bars | 126<br>126 | ~0.5y initial train (expanding)<br>扩展模式初始训练约半年 |
| test_bars<br>test_bars | 63<br>63 | ~1 quarter OOS per window<br>每窗约一季样本外 |
| step_bars<br>step_bars | 63<br>63 | Advance train/test frontier<br>训练/测试前沿步进 |
| rolling_train_bars<br>rolling_train_bars | 252<br>252 | 1y train when mode=rolling<br>滚动模式 1 年训练 |
| cost_assumption<br>cost_assumption | BASELINE<br>BASELINE | Costs applied inside each slice backtest<br>每片回测内应用成本 |

## 27.2 Algorithm
## 27.2 算法

- Build window bounds (train_start, train_end, test_start, test_end) exclusive end indices.
- For each window, run full `run_cross_sectional_backtest` on the price slice (rolling mode prepends warm-up bars so lookback signals are defined).
- Extract only OOS returns in [test_start, test_end); store per-window metrics.
- Concatenate OOS segments with de-duplication so overlapping steps do not double-count.
- Aggregate OOS Sharpe / return / drawdown from the de-duplicated OOS path — this is the promotion-relevant number, not in-sample Sharpe.

构建窗口边界（train_start…test_end），终点索引为开区间。

每窗对价格切片跑完整 `run_cross_sectional_backtest`（滚动模式前置预热 K 线，使回看信号有定义）。

仅抽取 [test_start, test_end) 的样本外收益；保存每窗指标。

拼接样本外片段并去重，避免步进重叠导致双重计数。

由去重后的样本外路径汇总 OOS 夏普/收益/回撤——这才是与晋升相关的数字，而非样本内夏普。

```
# engine/walk_forward.py (conceptual)
for (ts, te, vs, ve) in _window_bounds(n, wf_cfg):
    slice_prices = prices.iloc[:ve]          # expanding
    # rolling: include warm-up before ts
    bt = run_cross_sectional_backtest(slice_prices, cs_cfg, cost_model)
    oos = returns_in_original_index[vs:ve]   # exclusive end
    windows.append(WalkForwardWindow(..., metrics=calculate_metrics(oos)))
aggregate = calculate_metrics(dedupe_concat(oos_parts))
```

Interpretation: a strategy with IS Sharpe 1.8 but aggregate OOS Sharpe ≤ 0 is a failed candidate — regardless of narrative quality.

解读：样本内夏普 1.8 但汇总 OOS 夏普 ⩽ 0 的策略是失败候选——无论叙述多漂亮。

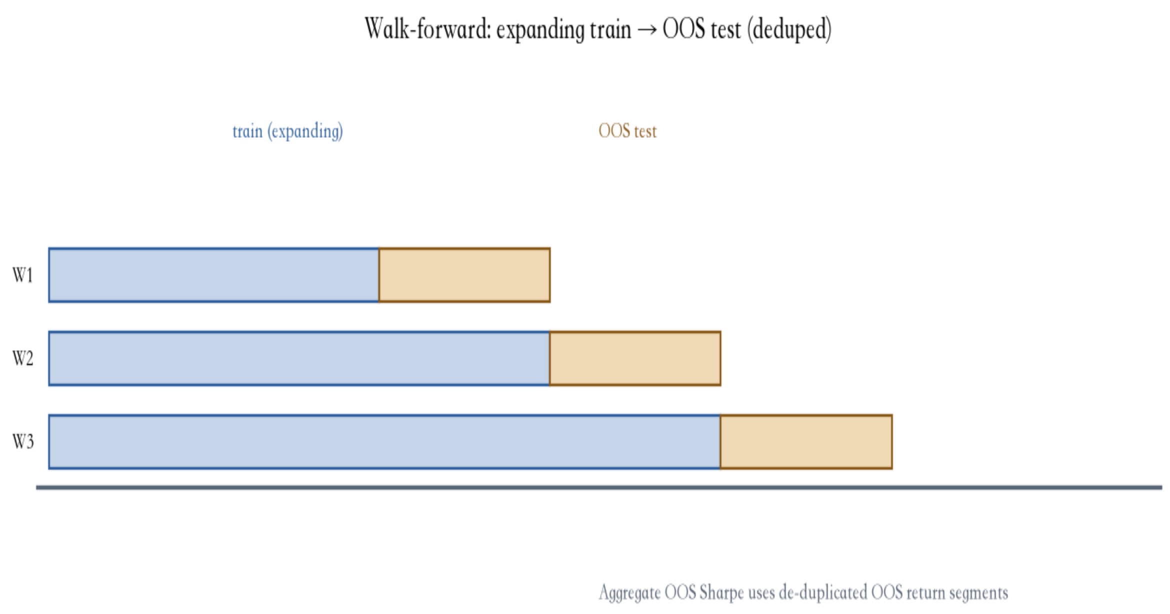


Figure 11. Expanding walk-forward train/test schedule.
图 11. 扩展式 walk-forward 训练/测试日程。

## 28. Deep Dive: Robustness & Fragility

## 28. 深入解析：稳健性与脆弱性

File: `engine/robustness.py` — `analyze_parameter_surface()`. The battery re-runs the cross-sectional backtest across a grid of one parameter (default: lookback ∈ {10,15,20,25,30,40,60}) and classifies the Sharpe surface.

文件：`engine/robustness.py` — `analyze_parameter_surface()`。套件在单一参数网格上重跑横截面回测（默认 lookback ∈ {10,15,20,25,30,40,60}），并对夏普曲面分类。

### 28.1 Fragility and smoothness rules

### 28.1 脆弱性与平滑规则

- **Fragile**: interior peak Sharpe exceeds neighbor mean by > 0.5 AND peak Sharpe > 0.3 → flagged as possible overfit knife-edge.
- **Boundary peak**: peak at first/last grid point → note to extend the grid before calling knife-edge.
- **Smooth**: mean |ΔSharpe| across adjacent grid points < 0.35.
- Preferred outcome: broad-ish stable region with positive Sharpe — not a single spike.

脆弱：内部峰值夏普高于邻域均值 > 0.5 且峰值 > 0.3 → 标记为可能过拟合的刀锋峰。

边界峰：峰在网格首/末点 → 提示先扩展网格再下刀锋结论。

平滑：相邻网格点平均 | Δ 夏普 | < 0.35。

更优结果：宽平台、正夏普区域——而非单点尖峰。

```
# Fragility test (interior peaks)
gap = peak_sharpe - mean(neighbor_sharpes)
fragile = (gap > 0.5) and (peak_sharpe > 0.3)
smooth = mean(abs(diff(sharpes))) < 0.35
```

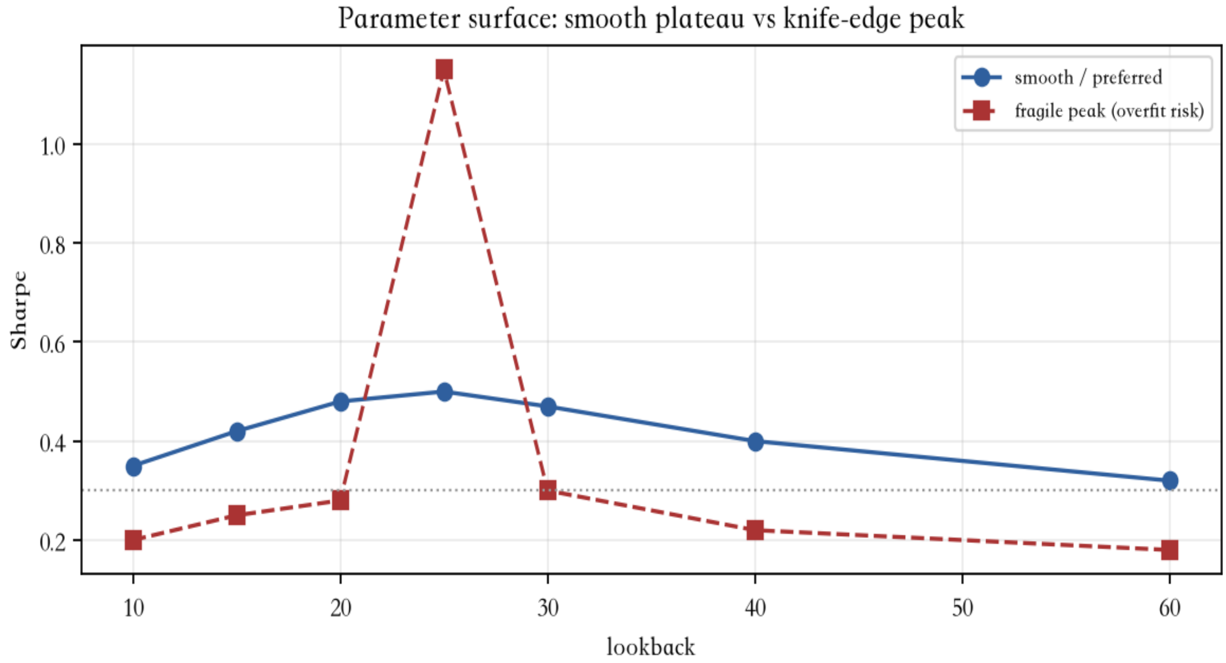


Figure 12. Illustrative smooth vs fragile lookback surfaces.
图 12. 平滑 vs 脆弱 lookback 曲面示意。

| | |
|---|---|
| Adversarial review treats fragile=True / jagged surfaces as promotion blockers even when the chosen lookback prints a high in-sample Sharpe. | 对抗性审查将 fragile=True / 锯齿曲面视为晋升阻断，即使所选 lookback 样本内夏普很高。 |

## 29. Deep Dive: Regime Analysis
## 29. 深入解析：市场状态分析

File: `engine/regime.py`. Labels are **heuristic** (rolling volatility tertiles × risk-on/off trend), not a claim of true latent states. Confidence is intentionally low (~0.4) so UI/report consumers do not over-trust the taxonomy.

文件：`engine/regime.py`。标签是启发式的（滚动波动率三分位 × 风险偏好趋势），并非真实隐状态声明。置信度故意偏低（~0.4），避免 UI/报告读者过度信任该分类。

- vol = rolling_std(market_proxy, 20); trend = rolling_mean(market_proxy, 20).
- vol_bin ∈ {low_vol, mid_vol, high_vol}; trend_bin ∈ {risk_on, risk_off}.
- Label string: `"{vol}|{trend}"`; shifted by 1 bar so day-t return is attributed to regimes known through t–1 (no same-day leakage).
- Per-regime metrics via pinned `calculate_metrics` (skip cells with < 5 observations).
- Concentration note when best regime Sharpe > 0.5 and worst < 0.

- vol = 市场代理 20 日滚动标准差；trend = 20 日滚动均值。
- vol_bin ∈ {low_vol, mid_vol, high_vol}；trend_bin ∈ {risk_on, risk_off}。
- 标签：`"{vol}|{trend}"`；平移 1 根 K 线，使第 t 日收益归因于截至 t 1 已知状态（无同日泄漏）。
- 各状态指标经固定 `calculate_metrics` 计算（观测 < 5 的单元跳过）。
- 当最佳状态夏普 > 0.5 且最差 < 0 时附加集中度备注。

| | |
|---|---|
| A strategy that only works in one historical regime slice is not book-ready — even if full-sample Sharpe looks fine. | 仅在单一历史状态切片有效的策略不具备入册条件——即使全样本夏普看起来不错。 |

## 30. Deep Dive: Diversification Mathematics
## 30. 深入解析：分散化数学

File: `alpha/registry.py` — `analyze_diversification()`. Flagship workflow compares candidates against seeded `ALP-existing-momentum`.

文件：`alpha/registry.py` — `analyze_diversification()`。旗舰流程将候选与种子 `ALP-existing-momentum` 比较。

| Output field 输出字段 | Definition 定义 | Promotion heuristic 晋升启发式 |
|---|---|---|

| | | |
|---|---|---|
| return_correlations<br>收益相关 | Pearson corr(candidate, existing_i) on aligned days<br>对齐日上 Pearson 相关 | Track max \|corr\| vs book<br>跟踪相对组合的最大 \|相关\| |
| downside_correlations<br>下行相关 | Corr on days where either series is negative<br>任一侧为负的日子上的相关 | Crash co-movement matters more than calm corr<br>崩盘共动比平静相关更重要 |
| genuine_diversification<br>真实分散 | \|avg corr\| < 0.35 AND \|avg downside corr\| < 0.45<br>\|平均相关\| < 0.35 且 \|平均下行相关\| < 0.45 | False → duplicate risk bet<br>False → 重复风险押注 |
| incremental_sharpe_hint<br>增量夏普提示 | Equal-weight existing vs existing+candidate Sharpe (ddof=0)<br>既有等权 vs 既有+候选 夏普（ddof=0） | delta > 0 preferred but not sufficient alone<br>delta > 0 更优但单独不足 |

```
# Diversification core
corr = aligned.c.corr(aligned.e)
down_mask = (aligned.c < 0) | (aligned.e < 0)
down_corr = aligned.loc[down_mask].corr()
genuine = abs(avg_corr) < 0.35 and abs(avg_down) < 0.45
port_old = existing_df.mean(axis=1)
port_new = concat([port_old, candidate], axis=1).mean(axis=1)
delta = sharpe(port_new) - sharpe(port_old)   # ddof=0 annualized
```

## 31. Deep Dive: ToolRouter & Allowlist

## 31. 深入解析：ToolRouter 与白名单

File: `tools/router.py`. Agents never call engine functions directly for research numbers. They invoke named tools through `ToolRouter`, which enforces an allowlist, records audit payloads, and returns structured `ToolResult` objects. This is the hard boundary that prevents LLM prose from inventing Sharpes.

文件：`tools/router.py`。智能体从不直接调用引擎函数获取研究数字。它们经 `ToolRouter` 调用具名工具；路由强制白名单、记录审计载荷，并返回结构化 `ToolResult`。这是阻止 LLM 叙述编造夏普的硬边界。

| Tool<br>工具 | Engine / store<br>引擎/存储 | Produces<br>产出 |
|---|---|---|
| create_experiment<br>create_experiment | SQLite experiments<br>SQLite experiments | experiment_id + configuration_hash<br>实验 ID + 配置哈希 |
| run_backtest<br>run_backtest | cross_sectional + metrics + costs<br>横截面+指标+成本 | backtest_id, returns, PerformanceMetrics<br>回测 ID、收益、指标 |
| walk_forward<br>walk_forward | walk_forward.py<br>walk_forward.py | OOS windows + aggregate<br>样本外窗 + 汇总 |
| robustness_battery<br>robustness_battery | robustness.py<br>robustness.py | surface, fragile, smooth<br>曲面、脆弱、平滑 |
| analyze_diversification<br>analyze_diversification | alpha/registry.py<br>alpha/registry.py | corr, genuine_diversification<br>相关、真实分散 |
| run_stress_test<br>run_stress_test | portfolio.py<br>portfolio.py | stress scenarios<br>压力情景 |
| run_bootstrap<br>run_bootstrap | statistics.py<br>statistics.py | uncertainty / multiple-testing aware stats<br>不确定性/多重检验感知统计 |

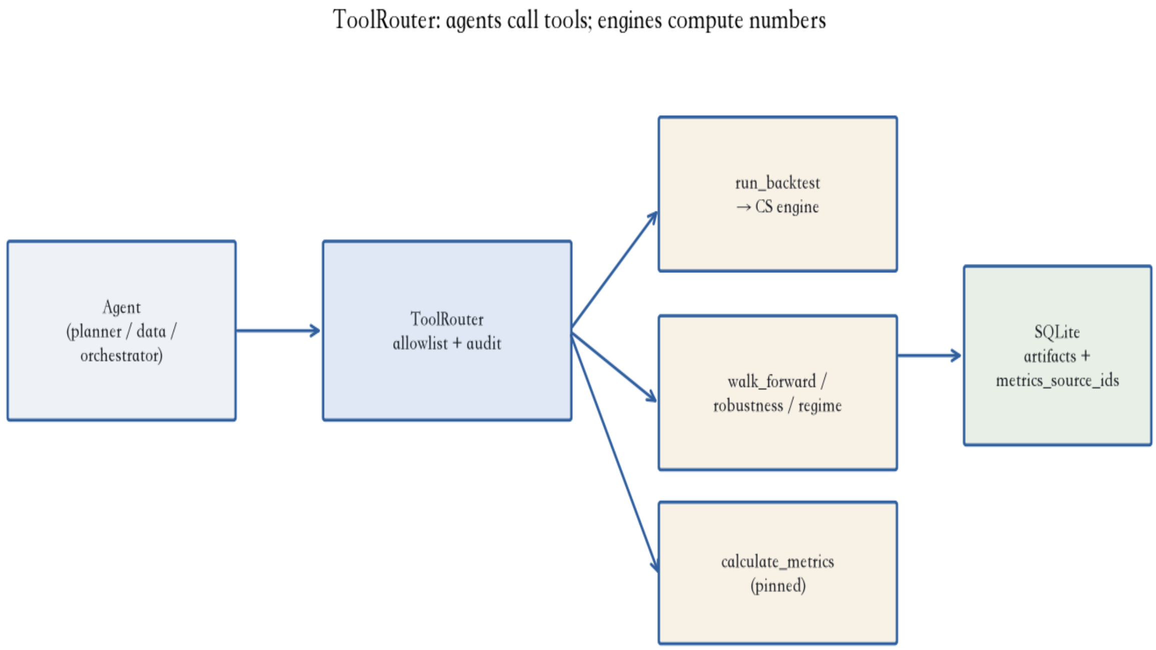


**Figure 13. Numbers flow Agent → ToolRouter → Engine → Storage.**
**图 13. 数字流：智能体 → ToolRouter → 引擎 → 存储。**

| | |
|---|---|
| Extension rule: new research capabilities must land as allowlisted tools with deterministic engines behind them — never as free-form model arithmetic. | **扩展规则：新研究能力必须以白名单工具落地、背后是确定性引擎——绝非自由形式的模型算术。** |

# 32. Worked Example: From Returns to Decision

# 32. 演算示例：从收益到决策

The following toy path illustrates how a researcher should mentally audit a candidate. Numbers are pedagogical (not a live run dump).

以下玩具路径说明研究员应如何心智审计一个候选。数字用于教学（非实况运行转储）。

## 32.1 Net daily returns → pinned metrics

## 32.1 净日收益 → 固定指标

Suppose a 5-day net return series after costs: `r = [+0.4%, −0.2%, +0.3%, −0.1%, +0.2%]`. Equity path: `1.004 → 1.0020 → 1.0050 → 1.0040 → 1.0060`. Cumulative ≈ +0.60%. With N=5 the annualization factor is noisy — in production N is hundreds of bars — but the formulas stay identical: `σ = std(r, ddof=0)·√252`, `Sharpe = mean(r)/std(r,ddof=0)·√252` (rf=0).

假设成本后 5 日净收益序列：`r = [+0.4%, −0.2%, +0.3%, −0.1%, +0.2%]`。权益路径：`1.004 → 1.0020 → 1.0050 → 1.0040 → 1.0060`。累计 ≈ +0.60%。N=5 时年化噪声大——生产中 N 为数百根 K 线——但公式不变：`σ = std(r, ddof=0)·√252`，`Sharpe = mean(r)/std(r,ddof=0)·√252`（rf=0）。

## 32.2 Cost stress

## 32.2 成本压力

If average one-way turnover per rebalance is 0.40 and rebalance_every=5, daily average turnover ≈ 0.40/5 = 0.08. Under BASELINE (2.75 bps variable) daily cost ≈ 0.08 × 2.75e-4 = 2.2e-5 (~0.22 bps/day). Under PESSIMISTIC (10 bps) ≈ 0.8 bps/day. A thin edge that disappears under this haircut is not promotable.

若每次再平衡单边换手 0.40 且 rebalance_every=5，日均换手 ≈ 0.40/5 = 0.08。BASELINE（2.75 bps 可变）下日成本 ≈ 0.08 × 2.75e-4 = 2.2e-5（约 0.22 bps/日）。PESSIMISTIC（10 bps）下≈ 0.8 bps/日。被这层摩擦吃掉的薄优势不可晋升。

## 32.3 Decision table

## 32.3 决策表

| Check<br>检查 | Pass criterion<br>通过标准 | Example outcome<br>示例结果 |
|---|---|---|
| IS Sharpe (audit only)<br>样本内夏普（仅审计） | Recomputable from returns<br>可由收益重算 | 1.1 — not decisive<br>1.1 — 非决定性 |
| OOS Sharpe<br>OOS 夏普 | > 0.3 after BASELINE costs<br>BASELINE 成本后 > 0.3 | 0.18 — fail<br>0.18 — 失败 |
| Fragility<br>脆弱性 | fragile=False, smooth≈True<br>fragile=False，smooth≈True | fragile=True — fail<br>fragile=True — 失败 |
| \|corr(momentum)\|<br>\|corr(动量)\| | < 0.35<br>< 0.35 | 0.62 — duplicate risk<br>0.62 — 重复风险 |
| Adversarial<br>对抗性 | severity < CRITICAL<br>严重度 < CRITICAL | HIGH cost sensitivity<br>HIGH 成本敏感 |
| Decision<br>决策 | PROMOTABLE / MORE_RESEARCH / REJECT<br>可晋升 / 需更多研究 / 拒绝 | REQUIRES_MORE_RESEARCH<br>REQUIRES_MORE_RESEARCH |

Honest outcome for the flagship synthetic FX demo is often REQUIRES_MORE_RESEARCH — that is a feature of scientific hygiene, not a product defect.

旗舰合成外汇演示的诚实结果常常是 REQUIRES_MORE_RESEARCH——这是科研卫生的特性，不是产品缺陷。

# 33. Workstation Page-by-Page Guide

# 33. 工作站逐页指南

The Next.js app under `web/` is the daily operator surface. Each page has one primary job. Use this map when onboarding or auditing a run.

`web/` 下的 Next.js 应用是日常操作面。每页只有一个主职责。入门或审计运行时用此地图。

| Route<br>路由 | Primary job<br>主职责 | Evidence to open next<br>下一步打开的证据 |
|---|---|---|
| /<br>/ | Fleet pulse: latest research, counts, health<br>总览：最新研究、计数、健康 | Click research → /research/[id]<br>点研究 → /research/[id] |

| | | |
|---|---|---|
| /research<br>/research | Queue of research requests + statuses<br>研究请求队列与状态 | Open workspace for a run<br>打开某次运行的工作区 |
| /research/[id]<br>/research/[id] | Workflow graph, plan, agents, report<br>工作流图、计划、智能体、报告 | Experiments sorted by Sharpe<br>按夏普排序的实验 |
| /experiments<br>/experiments | All experiments with enriched metrics<br>全部实验（富化指标） | Row → config hash + backtest<br>行 → 配置哈希 + 回测 |
| /alphas<br>/alphas | Library status + metrics_source_ids<br>库状态 + metrics_source_ids | Drill to source backtest<br>下钻到来源回测 |
| /portfolio<br>/portfolio | What-if book + Alpha X<br>情景：组合 + Alpha X | Risk page for concentration<br>风险页看集中度 |
| /risk<br>/risk | Correlation / concentration alerts<br>相关/集中度告警 | Reject high-corr adds<br>拒绝高相关新增 |
| /data<br>/data | Dataset catalog / synthetic panels<br>数据集目录/合成面板 | Confirm fx_synthetic_* labels<br>确认 fx_synthetic_* 标签 |
| /regimes<br>/regimes | Regime performance breakdown<br>状态绩效分解 | Check concentration notes<br>检查集中度备注 |
| /paper<br>/paper | Paper mode (IID honesty banners)<br>模拟盘（IID 诚实横幅） | Never treat as LIVE<br>绝不当作实盘 |
| /reports<br>/reports | Decision + survivors + narrative<br>决策 + 幸存者 + 叙述 | Traceability chain<br>可追溯链 |
| /agents<br>/agents | Allowlist / agent roles<br>白名单/角色 | Inspector payloads<br>检查器载荷 |
| /memory<br>/memory | Prior findings / memory store<br>既有发现/记忆库 | Avoid repeating failed ideas<br>避免重复失败想法 |
| /system<br>/system | API health, versions, ops<br>API 健康、版本、运维 | Restart / key config<br>重启/密钥配置 |

### 33.1 Research workspace mental model
### 33.1 研究工作区心智模型

- Graph nodes = orchestrator stages; green/completed means tools finished, not that alpha is good.
- Agent Inspector shows tool args/results — this is the audit log for numbers.
- Report decision is downstream of validation; always verify survivors' metrics_source_ids.
- Bottom command bar starts new research with budget (max_experiments) — budgets are first-class.

图节点 = 编排阶段；绿色/完成表示工具跑完，不表示 Alpha 优良。

智能体检查器显示工具参数/结果——这是数字的审计日志。

报告决策在验证下游；务必核幸存者 metrics_source_ids。

底部命令栏以预算（max_experiments）启动新研究——预算是一等公民。

## 34. Translation & Review Notes
## 34. 翻译与审阅说明

This section records terminology choices and a reviewer checklist for the bilingual edition.

本节记录双语版术语选择与审阅清单。

### 34.1 Terminology glossary
### 34.1 术语表

| English<br>英文 | Chinese (chosen)<br>中文（选用） | Notes<br>说明 |
|---|---|---|
| Alpha<br>Alpha | Alpha■■■■■■■■<br>Alpha | Keep "Alpha" as loanword<br>保留英文借词 |
| Sharpe ratio<br>Sharpe | ■■■■<br>夏普比率 | Standard CN finance term<br>标准金融译名 |
| Drawdown<br>Drawdown | ■■ / ■■■■<br>回撤 | Max DD → ■■■■<br>最大回撤 |
| Walk-forward<br>Walk-forward | ■■■■■■<br>滚动前瞻验证 | Also ■■■■<br>亦可称走步验证 |
| Look-ahead bias<br>Look-ahead | ■■■■<br>前视偏差 | Decision-time info only<br>仅决策时可得信息 |

| | | |
|---|---|---|
| Cross-sectional<br>CS | ■■■<br>横截面 | FX basket ranking<br>外汇篮子排序 |
| Adversarial review<br>Adversarial | ■■■■■<br>对抗性审查 | Critical review, not GAN<br>批评式审查，非生成模型 |
| Orchestrator<br>Orchestrator | ■■■<br>编排器 | Research state machine<br>研究状态机 |
| Deterministic<br>Deterministic | ■■■<br>确定性 | Opposite of stochastic LLM<br>相对随机 LLM |
| Paper trading<br>Paper | ■■■■■<br>模拟盘 | Labeled simulation mode<br>已标注的仿真模式 |
| Backtest<br>Backtest | ■■<br>回测 | Standard<br>标准译名 |
| Regime<br>Regime | ■■■■<br>市场状态 | Also ■■<br>亦可称区制 |
| Turnover<br>Turnover | ■■■<br>换手率 | Weight-space turnover<br>权重空间换手 |
| Provenance<br>Provenance | ■■ / ■■■■<br>溯源 | metrics_source_ids<br>指标来源 ID |
| Allowlist<br>Allowlist | ■■■<br>白名单 | Tool permit-list<br>工具许可名单 |

## 34.2 Reviewer checklist (this edition)
## 34.2 本版审阅清单

• No promotional exaggeration: paper mode and deterministic agents are disclosed in both languages.

• No inappropriate content; tone is institutional and technical.

• Quant terms use standard Mainland Chinese finance translations; English identifiers preserved.

• “Adversarial” means critical review of strategies (■■■■■), not generative-model jargon.

• Figures, captions, and tables are bilingual; code remains English with optional ZH comments.

• Side-by-side layout: English left column, Chinese right column with light background.

无夸大宣传：模拟盘模式与确定性智能体在两种语言中均如实披露。

无不当内容；语气保持机构化与技术性。

量化术语采用大陆常用金融译名；英文标识符保留。

“Adversarial”
指对策略的批评式审查（对抗性审查），而非生成模型术语。

图、图注与表格双语；代码保持英文并可选中文注释。

对照版式：左英右中，中文列浅底以利扫读。

| | |
|---|---|
| End of bilingual handbook. Optimize for researchers using the system for hours every day: traceability over spectacle, engines over narratives, budgets over unbounded loops. | 双语手册完。为每日长时间使用的研究员优化：可追溯优于炫技，引擎优于叙述，预算优于无界循环。 |